



AWS Amplify – Overview



What is AWS Amplify?

AWS Amplify is a set of tools and services that enables **developers to build scalable full-stack web and mobile apps** powered by AWS. It offers both **backend services** (like authentication, APIs, databases, and storage) and **frontend tools** (hosting and CI/CD).



Core Components of AWS Amplify

Component	Description
 Amplify CLI	Command-line tool to configure backend services (Auth, API, Storage, etc.)
 Amplify Libraries	Open-source client libraries for web & mobile apps to interact with backend
 Amplify Hosting	Fully managed CI/CD and hosting for web apps
 Amplify Studio	Visual development environment to build and manage app backend (no-code/low-code)
 Amplify UI	Ready-made React, Vue, Angular, and Flutter components (e.g., Auth UI)



What Can You Build With Amplify?

You can create **full-stack apps** with:

- **Authentication** (via Cognito)
- **GraphQL & REST APIs** (via AppSync + Lambda or API Gateway)
- **Databases** (DynamoDB or Aurora Serverless)

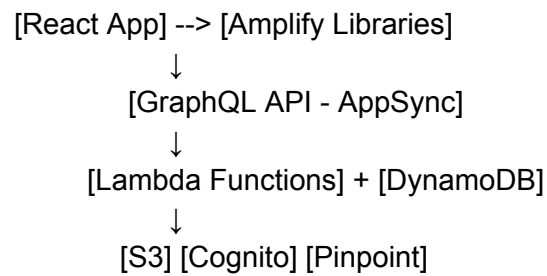
- **File Storage** (S3)
 - **Push Notifications**
 - **Custom Lambda functions**
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Supported Frontends

Amplify supports:

- **Web:** React, Angular, Vue, Next.js
 - **Mobile:** iOS (Swift), Android (Java/Kotlin), Flutter, React Native
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Typical Amplify Architecture



Common Use Cases

Use Case	Amplify Feature
User Sign-up/Login	Cognito + Auth UI
Upload images	S3 + Storage API
Real-time chat	AppSync + Subscriptions
Admin dashboard	Amplify Studio
Deploy React site	Amplify Hosting



Example: Build & Deploy a React App with Auth & API

Initialize Amplify

```
amplify init
```

1.

Add Authentication

```
amplify add auth  
amplify push
```

2.

Add GraphQL API

```
amplify add api  
amplify push
```

3.

Connect in React

```
import { API, Auth } from 'aws-amplify';
```

4.

Deploy

```
amplify publish
```

5.



Security Features

- IAM roles for resource-level access
- Cognito User Pools for user auth

- OAuth2.0, SAML, Social login (Google, Facebook, etc.)
 - AppSync & GraphQL fine-grained access control
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Pricing

- **Amplify Hosting:** Free tier + pay-as-you-go (\$0.01 per build minute, \$0.023/GB stored/month)
 - **Backend Services:** Based on usage (Cognito, AppSync, Lambda, S3, etc.)
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Pros

- ✓ Easy integration with AWS services
 - ✓ Frontend and backend managed together
 - ✓ Scales automatically
 - ✓ Developer-friendly CLI & libraries
 - ✓ Built-in CI/CD
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Cons

- ✗ Limited control over infrastructure (compared to raw CDK/IaC)
 - ✗ Costs can scale with usage
 - ✗ Complex migrations in large apps
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Ideal For

- Startups & MVPs
- Rapid frontend-backend prototyping
- Full-stack serverless application